

AAVAR KHATIWODA

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Falls Church, VA | Active Secret Clearance

SUMMARY

Versatile software engineer with a wide breadth of experience in diverse technologies, from web development, scripting, systems engineering, and memory compression techniques. Proven track record of leading research teams, spearheading research execution, and upkeeping critical systems.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C++, C, Bash, RISC-V Assembly

Frameworks and Tools: ReactJS, JSX, Sass, Linux / Red Hat Enterprise Linux, Virtual Machines, Terminal, Vim, Git, Jenkins, Jira

Operating Systems & Memory: Compression and Random-Access Decompression Techniques in Memory

PROFESSIONAL HISTORY

Lockheed Martin

Software Engineer

May 2024 - Present

Manassas, VA

- Provide full-lifecycle development, implementation, and maintenance of robust and scalable internal/client applications, and extend functionality of legacy tools.
- Lead new project of 10+ engineers and develop data transmission pipelines from specialized hardware to dedicated systems. Coordinate project architecture planning, and development story writing/execution over 6+ months. Configure network hardware and validate traffic through analyzing packet capture and replay.
- Promote system maintenance and uptime by developing and integrating Bash scripts to log and export critical tool output to aid in expedited detection and analysis of failing system processes. Promote robustness and modular scalability of system functions.

Northrop Grumman

Software Engineer Intern

Jun 2023 - Aug 2023

Baltimore, MD

- Led team to achieve above-expected story completion targets through effectively coordinating resources and prioritizing high-risk stories as rotating scrum lead.
- Developed new health monitoring scripts used on the Jenkins pipeline to better diagnose and prevent impending build failures, implementing diagnostic filtering of health data on OpenSearch Dashboards.

RESEARCH

Virginia Tech, College of Engineering

Undergraduate Research Assistant, CS 4994 - Undergraduate Research

Jan 2024 - May 2024

Blacksburg, VA

- Resolved memory bandwidth limitations of traditional LZ77 and Huffman compression techniques through designing a novel 3-step algorithm tailored to specifications of memory, improving compression and random-access decompression across 4 large and diverse memory dumps.
- Outperformed the DEFLATE algorithm in compression performance across each tested memory dump.

TECHNICAL PROJECTS

Software Engineer Portfolio

aavar.studio

2020 - Present

Continuous Learning & Development

- Self-taught ReactJS developer with 5+ years of experience. Built custom professional software engineer portfolio page to highlight relevant technical expertise and leadership/research experience.
- Proficient in UI development with React. Designed fully custom and responsive layouts for common screen sizes up to 4k resolution.

LEADERSHIP

CS OpenSource at Virginia Tech, Career Development Resources

Founding Vice President

Jan 2024 - May 2024

Blacksburg, VA

- Established a new career development-focused club targeting students' understanding of AGILE software development.
- Introduced 55+ new students to the software development lifecycle and critical workflow tools [eg., AGILE, git, terminal].

Virginia Tech, College of Engineering

Undergraduate Teaching Assistant, CS 1114 - Introduction to Software Design

Aug 2023 - Dec 2023

Blacksburg, VA

- Aided 200+ college students in building a critical understanding of object-oriented programming concepts in Java.

EDUCATION

Virginia Tech, College of Engineering

Bachelor of Science in Computer Science

Minor in Mathematics

May 2024

Blacksburg, VA

Dean's List with Distinction